Designing LLM-Driven Agents Across Individual and Organizational Intelligence Levels

A SARSI-worker-based engineering framework for I0-IΩ agents, O0-OΩ organizations, and circle-indexed recursive self-improvement

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Abstract

This paper gives an engineering design for LLM-driven agents at each individual intelligence level (I0 reactive through IΩ open-ended self-evolving) and each organizational intelligence level (O0 coordinated through OΩ open-ended evolving organization). The central claim is that an LLM does not by itself determine an agent's level. A level is produced by the surrounding architecture: persistent state, write set, working-memory gate, evidence ledger, learning loop, independent evaluator, promotion authority, and organizational separation. The SARSI-worker design is used as the common cognitive chassis: it supplies observations, episodic memory, semantic memory, procedural memory, intentional state, self-state, world/task model, mutable cognitive instruments, recent dialogue state, and a bounded Gate_B context assembler. The paper shows how I1 and I2 arise directly from the current SARSI-worker plan, how I3 and I4 require governed Theta and meta-improver modification, how I5 adds autonomous discovery, and how IΩ adds discovery-to-cognitive-tool transduction with a frozen referee. Organizational levels are then built by coordinating multiple differently constrained SARSI-style individuals together with external verifiers and promoters. The result is a practical blueprint for building agent systems whose intelligence level is measured by valid persistent change rather than by raw model capability or self-description.

1. Introduction

Large language models can reason, write code, explain scientific ideas, and operate tools. But this does not automatically make an LLM an evolving agent. An individual or organizational intelligence level is not a property of the model alone. It is a property of a complete system: the model, its persistent state, what it may modify, what evidence it records, who evaluates its changes, and who has authority to promote those changes.

The design problem is therefore not: 'Which prompt makes the LLM IΩ?' The design problem is: 'What architecture lets an LLM-driven system validly remember, learn, self-improve, discover, and evolve while preserving independent evaluation and authority boundaries?' This paper answers that design question.

The paper uses the current SARSI-worker BrainRSI v3 plan as the individual-agent chassis. That plan treats sarsi-worker as the persistent brain, sarsi-claude as a transient motor executor, and an independent mechanism as the verifier. It also introduces a mode-aware fast path so ordinary chat does not pay the cost of the full action pipeline. The same chassis can be configured into different individual levels, and networks of such individuals can be composed into organizational levels.

2. Core principle: LLM is an engine, not a level

An LLM is the generative and reasoning engine inside the agent. The agent level is produced by the surrounding system. The same base model can instantiate a low-level executor, a persistent assistant, an adaptive learner, a self-improving cognitive system, or a scientific discovery agent depending on what state it can persistently modify and how its changes are evaluated.

```
LLM + state + memory + tools + evidence + evaluation + promotion + governance = agent system
```

This leads to a design rule:

Never classify an agent level from model capability alone. Classify it from the validated persistent change loop it can perform.

3. The SARSI-worker chassis

The SARSI-worker core can be used as the base class for most individual agents above I0. Its target state can be summarized as:

```
Sigma_worker = {
  O,      # observations and evidence references
  E,      # episodic memory
  S,      # semantic memory
  P,      # procedural memory
  I,      # intentional memory: goals, plans, phase, checkpoints
  Q,      # self model: authority, calibration, health, freshness
  M,      # world/task model
  Theta   # mutable cognitive instruments: gate, retrieval, learning policies
}

R_t    = bounded recent-dialogue buffer / discourse state
Mode_t = CHAT | REASON | ACTION

W_t = Gate_B(O_t, R_t, I_t, protected_directives, Q_t,
             retrieved_S, retrieved_E, applicable_P, mode=Mode_t)
```

The most important engineering feature is that Sigma_worker is not sent wholesale into the model. Gate_B creates a bounded working context W_t. In CHAT mode, the context is small and conversational. In REASON mode, it retrieves relevant long-term memory. In ACTION mode, it additionally requires protected directives, readiness checks, prediction, execution, and verification.

Mode	Purpose	Default design
CHAT	Ordinary conversation, explanation, local follow-up	Use current turn + recent dialogue + minimal identity. No executor, forecast, deep retrieval, or verifier by default.
REASON	Conceptual reasoning, explicit recall, planning discussion with no side effect	Use current turn + recent dialogue + retrieved semantic/episodic memory + relevant intention/self-state.
ACTION	Create, assign, edit, run, commit, deploy, or other consequential operation	Use protected directives, operation-specific readiness, forecast, executor path, deterministic or independent verification, and checkpoint update.

4. Write sets, authority, and validity

An agent's level is determined largely by its write set. The write set is the part of the persistent brain the agent may modify. A memory-only agent is very different from an agent that can change its retrieval policy, skill library, objective, or weights.

Depth	Writable substrate	Design meaning
D0	Nothing persistent	Reactive or evaluator role. Useful for independent judgement because it cannot improve itself by its own verdict.
D1	Memory M	Episodes, notes, and retrieved context may persist.
D2	Policy Pi	Prompt profiles, thresholds, routing weights, budgets, and playbook parameters may change.
D3	Skills Sigma	New tools, workflows, procedures, subagents, or scaffolds may be created.
D4	Weights theta	Adapters, fine-tunes, or weight updates may be prepared, but deployment still requires independent evaluation and promotion.

The central safety and scientific-validity rule is:

```
Evaluator not in WriteSet
```

If the same agent can modify the mechanism and also declare that the modification improved it, the improvement claim is not meaningful. Therefore proposer, evaluator, and promoter should be separate for every level that makes learning or improvement claims.

5. Agent manifest: one codebase, many levels

Rather than building separate programs for I0, I1, I2, and so on, define a common agent runtime and instantiate it by manifest. The manifest states which memories exist, what the agent can write, which loops are enabled, and which external services evaluate and promote changes.

```
agent_id: research-worker-01
base: sarsi-worker
modes: [CHAT, REASON, ACTION]

state:
  episodic_memory: true
  semantic_memory: true
  procedural_memory: true
  intentional_memory: true
  self_model: true
  world_model: true
  theta: true
```

```
write_set:
  memory: true
  policy: candidate_only
  skills: candidate_only
  theta: candidate_only
  objective: false
  weights: false

learning:
  consolidation: true
  proceduralization: true
  held_out_ablation_required: true

rsi:
  theta_modification: false
  meta_improvement: false

discovery:
  unknown_model: false
  hypothesis_engine: false
  experiment_planner: false

authority:
  may_execute_tools: allowlisted
  may_modify_workspace: false
  may_deploy_self_change: false

verifier:
  external_required: true
  promoter_required: true
```

A different manifest can instantiate a D0 verifier, an I2 specialist, an I3 self-improving worker, or an I5 discovery agent. The LLM can help design these agents, but permissions should be enforced by infrastructure: file-system permissions, database ACLs, immutable ledgers, separate services, container boundaries, and owner-signed promotion.

6. Individual agent designs: I0 through IΩ

Level	Definition	Writable state	Design modules	Transition test
I0 Reactive agent	Responds from current input and ephemeral context; no persistent evolving cognitive state.	None, except audit logs.	Prompt, LLM, optional fixed tools, no writable long-term memory.	Restart erases learned facts. The same input under the same configuration behaves as a fresh invocation.
I1 Persistent agent	Maintains durable identity, task state, memory, goals, and conversation history.	Episodic memory, semantic references, intentional state, recent dialogue buffer, self-state.	sarsi-worker M1/M2: stable ids, append-only logs, Gate_B, R_t, current plan/checkpoint, active directives.	After restart, it resumes the right task and retrieves relevant prior decisions without loading the entire history.
I2 Adaptive-learning agent	Experience changes future behavior through fixed learning mechanisms.	Episodes, semantic lessons, procedural memory, scoped policy parameters if validated.	M5: trigger -> episode -> candidate lesson -> consolidation -> evidence check -> promotion.	Held-out matched tasks improve with learned memory/procedures compared with memory absent. Memory growth alone is not enough.
I3 Self-improving agent	Uses evidence about its own performance to change a cognitive mechanism.	Theta candidates: Gate_B policy, retrieval weights, planner, context budgeter, tool-selection policy, skill-generation rules.	Post-M5 governed RSI: failure diagnosis, candidate Theta', replay, challenge tests, canary, independent verifier, promoter.	Theta_new persists and outperforms Theta_old on predeclared evaluations without weakening authority or benchmarks.
I4 Recursive self-improving agent	Improves the mechanism that generates and evaluates future self-improvements.	Psi: failure clustering, candidate generation, meta-search, evaluation-budget allocation, challenge generation.	Meta-RSI loop: Improver-v1 -> evidence -> Improver-v2 -> independent measurement of Improvement Competence.	Improvement Competence increases: validated downstream gain per proposal/evaluation cost rises under comparable tasks.
I5 Autonomous-discovery	Identifies significant	Structured unknowns,	Add discovery layer:	Produces externally validated

agent	unknowns, formulates hypotheses, obtains evidence, and validates new knowledge.	hypotheses, experiments, knowledge graph, world model, scientific procedures.	unknown detector, question generator, hypothesis engine, experiment planner, evidence acquisition, scientific verifier.	knowledge not contained in its prior state or prompt; novelty is not enough without evidence.
IΩ Open-ended self-evolving individual	Discoveries reorganize the agent and become new cognitive tools that expand its future discovery space.	Knowledge-to-cognitive-tool transduction; new representations, new procedures, new Theta/Psi candidates, objective revisions under frozen referee.	I5 + cognitive transducer + versioned frozen referee + recursive tool/representation creation.	A validated discovery creates a cognitive tool that enables a later discovery class previously inaccessible to the agent. Objective revisions are scored against retained predecessor referees.

6.1 I0: reactive executor

I0 is best designed as a lightweight LLM invocation or tool wrapper. It should not carry the entire sarsi-worker memory substrate. It is useful for low-cost, high-volume work: parsing, formatting, deterministic transformation, running tests, extracting fields, or executing mature procedures. I0 agents are also useful as independent evaluators if their write set is empty.

6.2 I1: persistent SARSI-worker

I1 is the first natural home of sarsi-worker. It needs durable identity, episodic history, semantic references, intentional task state, and a bounded working-memory gate. The goal is continuity, not learning. An I1 worker can remember that a plan exists and resume it; it need not yet prove that historical experience improves future performance.

6.3 I2: adaptive SARSI-worker

I2 corresponds closely to BrainRSI v3 Phase M5. A machine-observable trigger creates an episode. Episodes may generate candidate semantic lessons or procedures, but a lesson does not become active merely because it was generated. It must pass evidence checks, avoid contradiction, and preferably improve held-out future tasks.

```
trigger -> episode -> candidate lesson -> consolidation -> evidence check -> semantic
promotion
repeated success -> procedure candidate -> sandbox tests -> active skill
```

6.4 I3: governed Theta-modifying agent

I3 begins only after the system has enough replayable evidence. It modifies Theta, not merely memory. Candidate changes might adjust Gate_B budgets, retrieval ranking, planning policies, or consolidation thresholds. The agent can propose such changes, but a separate verifier and promoter decide whether they become active.

6.5 I4: recursive improver

I4 makes the self-improvement mechanism itself a measured object. If the improver becomes better at proposing useful changes at lower cost, the system has recursive self-improvement. The relevant metric is not raw task score; it is Improvement Competence: validated downstream gain divided by proposal and evaluation cost.

6.6 I5 and IΩ: discovery and open-ended cognitive evolution

I5 requires structured ignorance: the agent must represent what is unknown, uncertain, contradictory, assumed, and experimentally inaccessible. IΩ goes further: validated discoveries become new cognitive tools or representations that alter what the agent can later discover. This is where scientific discovery and BrainRSI converge.

7. Organizational agent designs: O0 through OΩ

An organization is not one larger LLM. It is a set of individuals plus coordination structure: who proposes, who evaluates, who promotes, who holds authority, which evidence is shared, and how the organization couples to the world. Raw private memory should not be copied across agents. Cross-agent communication should be evidence-bearing.

```
message = {
  claim: "...",
  evidence_refs: [...],
  provenance: "agent/task/run/version",
  scope: "...",
  status: "candidate|verified|rejected|superseded"
}
```

Level	Definition	Core modules	Transition test
O0 Coordinated	Routes work among individuals, no persistent organizational learning.	Router, task queue, result collector.	After restart, no learned allocation or shared organizational state is required.
O1 Persistent	Maintains shared organizational memory, roles, task history, evidence, and decisions.	Agent registry, role registry, evidence index, task ledger, organizational memory.	Organization can recover who did what, what evidence exists, and what remains unresolved.
O2 Adaptive	Collective history changes future allocation or procedures through fixed learning mechanisms.	Routing analytics, performance records, workflow preference model, separated evaluator/promoter.	Future routing improves on held-out task families; proposer/evaluator/promoter are distinct.
O3 Self-improving	Organization modifies its own routing, memory, communication, verification, or role structures.	Org-Theta candidates, replay/shadow deployment, canary organizational tests.	Org structure v2 outperforms v1 under independent evaluation.
O4 Recursive self-improving	Organization improves how it discovers and validates future organizational improvements.	Org-Psi meta-improver, policy-search engine, challenge generation, budget allocation.	Organizational Improvement Competence rises under comparable evaluation conditions.
O5 Autonomous-discovery organization	The organization collectively discovers validated new knowledge through coordinated roles no single member fully contains.	Research agenda, theorists, experiment designers, operators, instrument makers, independent referees.	Discovery requires organizational structure and passes external/empirical validation.
OΩ Open-ended evolving organization	Discoveries create new tools, roles, agents, and institutional structures that expand future collective discovery.	O5 + organizational transducer + role/agent generator + frozen/referee governance.	New organizational forms created from discoveries open previously inaccessible problem classes.

7.1 The separation gate

From O2 upward, organizational claims are improvement claims. Therefore proposer, evaluator, and promoter must be distinct loci, and the evaluator's criteria and evidence must lie outside the proposer's write set. Otherwise the correct classification is not O3 or O4; it is O1 plus unverified claims of higher behavior.

7.2 Heterogeneity is required

A mature organization should not be made entirely of maximum-level individuals with identical authority. Referees and promoters often need lower self-write capability precisely because their trust comes from being outside the candidate's mutable cognition. High-level agents discover and redesign; middle levels adapt and proceduralize; low-level agents execute mature procedures cheaply and reproducibly.

8. Role design matrix

Role	Recommended individual level	Why this level fits	Main anti-pattern
Executor	I0-I1	Cheap, reliable performance on mature procedures.	Using IΩ for routine execution.
Persistent operator	I1	Keeps task state and evidence history.	Treating memory as proof of learning.
Adaptive specialist	I2	Learns from repeated task families.	Promoting lessons without held-out evidence.
Cognitive-method improver	I3	Improves Gate_B, retrieval, planning, skill selection.	Letting the proposer self-certify improvement.
Meta-improver	I4	Improves the improvement engine itself.	Measuring only task score instead of Improvement Competence.

Scientist/discoverer	I5	Finds unknowns and validates new knowledge.	Calling novelty or speculation discovery.
Frontier cognitive innovator	IΩ	Turns discoveries into new ways of thinking.	Objective drift without frozen referee.
Independent referee	D0 or tightly restricted	Trust comes from being outside the write set.	Using the same agent that proposed the change.
Promoter	D0 or limited other-reach	Activates versions only after verified evidence.	Letting candidate systems deploy themselves.
Director/principal	Objective-open but governed	Sets or revises objectives under frozen criteria.	Writing the evaluator after seeing the candidate.

9. Circle-indexed deployment

The same I/O architecture can operate inside each SARSI-L substrate circle. The circle tells us what substrate is recursively improved; the I-level tells us what kind of cognition an individual performs; the O-level tells us how multiple individuals make valid collective progress.

Circle	Substrate	Typical high-level agents	What D3 instrument-makers build	Dominant verifier
Circle I	Software: architecture, training process, weights, tools	I3/I4 workers; I5 computing researchers	Gate_B changes, retrieval systems, planners, skills, model-update scaffolds	Held-out software/evaluation benchmarks, deterministic tests, replay
Circle II	Hardware: chip design and fabrication	I5 hardware researchers; I3 EDA improvers	EDA tools, chip-design workflows, process-control tools	Simulation, physical chip measurements, yield and performance tests
Circle III	Physical: matter, machines, self-repair, self-replication	I5 materials/robotics agents; I3 control/manufacturing improvers	Robots, sensors, machine tools, repair/manufacturing procedures	Physical world response, durability, repair, replication tests
Circle IV	Biological: mechanisms, therapies, aging, clinical response	I5 biological discovery agents; IΩ scientific-method agents	Assays, experimental platforms, therapeutic workflows, analysis instruments	Empirical biology, replication, safety and efficacy endpoints
Circle V+	Stellar: orbital manufacturing and energy capture	Distributed I5/IΩ individuals inside OΩ structures	Orbital collectors, autonomous construction systems, resource pipelines	Net-growth of productive infrastructure after losses and maintenance

The critical point is that Circle level is not another intelligence ladder. A system could have strong I-level agents and weak circle closure if it lacks physical instruments. Conversely, a civilization could capture large energy resources while still having poor organizational separation or weak discovery capability.

10. Implementation roadmap using SARSI-worker phases

The uploaded BrainRSI v3 plan is best interpreted as the I1-to-I2 foundation. It deliberately stops before governed recursive self-improvement. Therefore higher levels should be added only after persistence, replay, prediction, verification, and consolidation produce longitudinal evidence.

Stage	SARSI phase	Resulting capability	Approximate level
Baseline	M0	Reproducible targeted test baseline and documented instability.	Prerequisite
Persistent contracts	M1	Durable ids, append-only directives, semantic/episode ledgers.	I1 substrate
Gate/retrieval/replay	M2	Mode-aware CHAT/REASON/ACTION, Gate_B, exact context replay.	I1 strong
Forecast/plan executive	M3	Pre-action expectations, calibration, restartable plans.	I1/I2 support
Executor/verifier loop	M4	sarsi-claude artifacts judged by independent criteria.	I2 prerequisite
Learning/consolidation	M5	Episodes become validated semantic/procedural candidates.	I2
Governed Theta RSI	M6	Cognitive mechanism candidates are replayed, tested, and promoted.	I3
Meta-RSI	M7	Improvement engine itself becomes measurable and updatable.	I4

Discovery layer	M8	Unknowns, hypotheses, experiments, and validated new knowledge.	I5
Transduction layer	M9	Discoveries become new cognitive tools and representations.	IΩ

11. Required services for a serious implementation

The design should be implemented as separated services rather than one all-powerful process. This matters because organizational level depends on separation, and individual self-improvement depends on external evaluation.

Service	Purpose	Must not be writable by
Agent runtime	Runs LLM, Gate_B, local memory, planning, and tool calls.	Itself for protected governance paths.
Memory service	Stores episodes, semantics, procedures, directives, context snapshots.	Unauthorized agents; old rows must not be rewritten.
Candidate store	Stores proposed policy/tool/Theta/organization changes.	Evaluator cannot mutate candidates during judging.
Evaluation service	Runs held-out tests, deterministic criteria, scientific verification.	Candidate proposer.
Promotion service	Activates new baselines only after evidence and authority checks.	Candidate proposer and runtime task executor.
Authority kernel	Defines permissions, credentials, grants, protected paths.	All mutable cognition.
Audit ledger	Immutable event/provenance chain.	All agents except append-only writer with checks.

12. Metrics and falsifiability

Do not create one brain score. Each level requires its own measurable transition evidence.

Claim	Minimum measurement
I1 persistence	Restart recovery, correct task/phase continuation, relevant memory retrieval, irrelevant memory omission.
I2 learning	Held-out future performance improves because of learned memory/procedure, not because of replayed instructions.
I3 self-improvement	Candidate Theta change outperforms frozen prior Theta under external evaluation and survives rollback tests.
I4 RSI	Improvement Competence rises: more validated downstream benefit per proposal/evaluation cost.
I5 discovery	New externally validated knowledge not contained in prior state or prompt.
IΩ open-ended cognition	Discovery creates a new cognitive tool that enables a later previously inaccessible discovery class.
O2+ organization	Separation holds: proposer, evaluator, promoter are distinct; organizational behavior improves on held-out outcomes.
OΩ organization	New knowledge produces new tools/roles/structures that expand collective discovery reach.

13. Risks and failure modes

- Level inflation: calling a powerful LLM I3 or I4 without evidence of persistent self-modification or improvement competence.
- Memory inflation: treating more stored information as learning even when future performance does not improve.
- Self-certification: allowing the proposer to judge or promote its own change.
- Authority drift: letting cognitive improvement widen permissions, benchmark ownership, or promotion rights.
- Objective drift: allowing IΩ-like objective evolution without a versioned frozen referee.
- Organizational monoculture: filling all roles with identical high-level agents and losing separation.

- Overuse of heavy machinery: forcing simple chat through forecast, executor, and verifier pipelines, making the system worse to use.
- Semantic authority error: using vector similarity to decide hard constraints instead of scope-based protected directives.

14. Conclusion

Yes, LLMs can design and drive agents at every individual and organizational level, but only when embedded in the right architecture. The LLM proposes, reasons, summarizes, and generates candidate actions. The agent level comes from persistent state, write depth, evidence, evaluation, promotion, and governance. SARSI-worker provides a strong chassis for I1 and I2 today: persistent memory, bounded Gate_B context, mode-aware conversation, replay, prediction, verification, and consolidation. Governed Theta modification can extend it to I3; meta-improvement can extend it to I4; scientific unknown/hypothesis/evidence loops can extend it to I5; and discovery-to-cognitive-tool transduction under a frozen referee can extend it toward IΩ.

Organizational levels are built not by making one agent larger, but by coordinating many differently constrained agents with evidence-bearing communication, independent verifiers, separate promoters, and world-coupled validation. The mature system should therefore contain all levels simultaneously: low-level executors, persistent operators, adaptive specialists, self-improving method agents, recursive improvers, discovery agents, frontier cognitive innovators, and low-write independent referees. The result is an engineering path from LLM-driven assistance toward valid, measured, recursive intelligence.

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